see at the detector is a pattern of individual points where we detect the particles, arranged in more or less a slit-like pattern.

We could also do the same thing with waves, for example, by placing the slit in a tub of water and creating waves that pass through it. When the waves pass through, they spread out in a semicircular pattern before eventually reaching the screen. Of course, we don't observe particle-like points when the water wave hits the screen, but let's imagine we have a special screen that lights up with a brightness that depends on the amplitude the waves reach at any particular point. They will be brightest at the point of the screen that is closest to the slit, and gradually fade as we get farther away.

Now let's do the same thing, but with two slits in the way rather than just one. The particle case isn't that much different; as long as our source of particles is sufficiently random that particles pass through both slits, what we'll see on the other side is two lines of points, one for each slit (or one thick line, if the slits themselves are sufficiently close together). But the wave case is altered in an interesting way. Waves can oscillate downward as well as upward, and two waves oscillating in opposite directions will cancel each other out—a phenomenon known as *interference*. So the waves pass through both slits at once, emanating outward in semicircles, but then set up an interference pattern on the other side. As a result, if we observe the amplitude of the resultant wave at the final screen, we don't simply see two bright lines; rather, there will be a bright line in the middle (closest to both slits), with alternating dark/bright regions that gradually fade to either side.